

CSIT302 Cybersecurity

Day 4-2 – Privacy Issues

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School of Computing and Information Technology

Outline

- Data Privacy Problem
- Approaches for Protecting Data Privacy
- Differential Privacy:
 - Definition and Design Mechanisms

Data Privacy: The Problem

- Given a dataset with some sensitive information, such as

- Census data
- Health records
- Social network activity
- Telecommunication data

- Academic Research
- Informing Policy
- Market Analysis
- ...

- How can we:

- enable “**desirable use**” of the data, such as
- while protecting the “**privacy**” of the data subjects?

What is the point?

Approach 1: Encrypt the Data

Name	Sex	Blood	...	HIV?
Chen	F	B	...	Y
Jones	M	A	...	N
Smith	M	O	...	N
Ross	M	O	...	Y
Lu	F	A	...	N
Shah	M	B	...	Y



Name	Sex	Blood	...	HIV?
100101	001001	110101	...	110111
101010	111010	111111	...	001001
001010	100100	011001	...	110101
001110	010010	110101	...	100001
110101	000000	111001	...	010010
111110	110010	000101	...	110101

Problems?

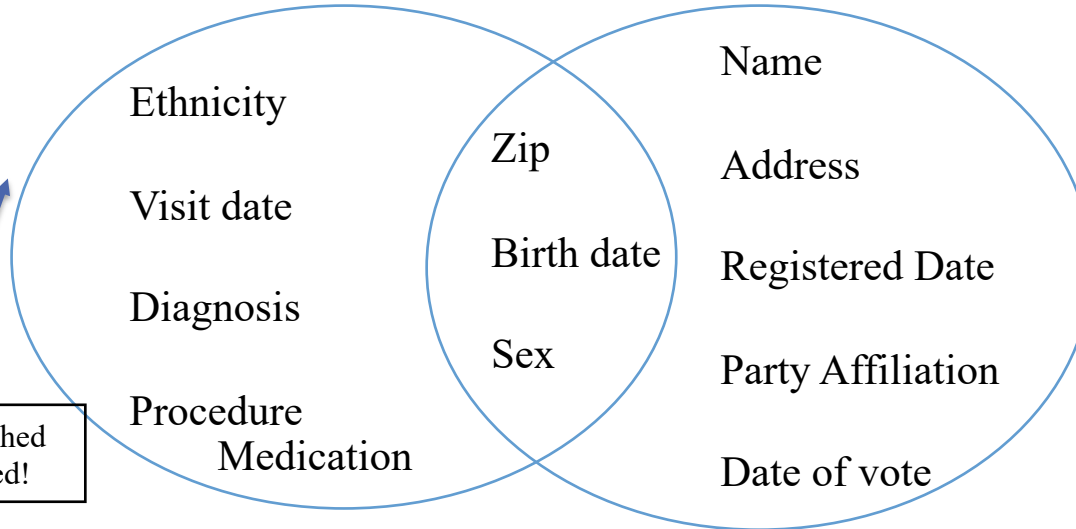
Approach 2: Anonymize the Data

No.	Sex	Blood		HIV?
1	F	B		Y
2	M	A		N
3	M	O		N
4	M	O		Y
5	F	A		N
6	M	B		Y

Problem?

Re-identification via Linkage

- Massachusetts Group Insurance Commission (late 1990s)



Latanya Sweeney
Re-identification Attack
(1997)

Data can be matched
up and reidentified!



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Anonymized Medical Data

Public Vote Data

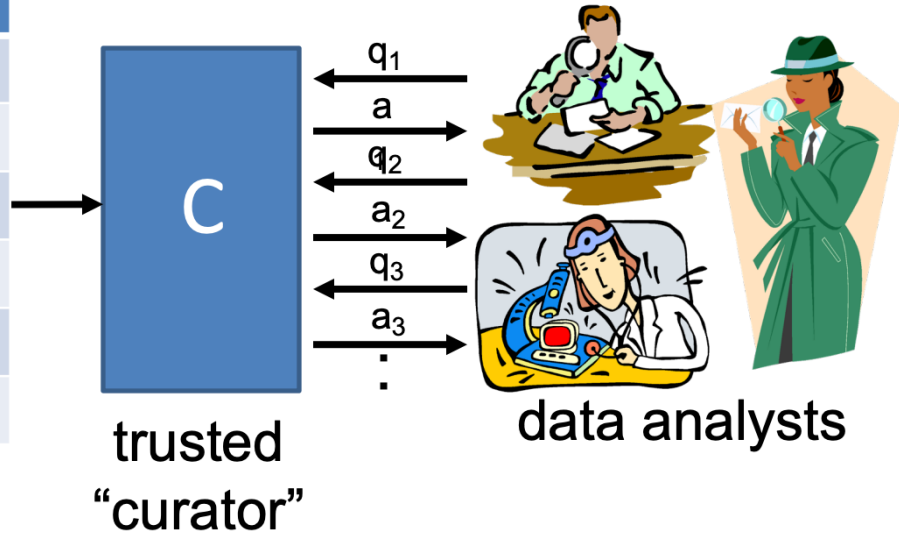
Netflix Prize

- Deanonimize movie ratings of anonymous Netflix subscribers
 - In 2006, Netflix opened a competition with \$1M award for predicting user's preference in movies.
 - It released some anonymized dataset consisting of user ID, movie ID, rates date.
 - Narayanan and Shmatikov matched users with IMDB dataset
 - Cancellation of the 2nd Netflix prize.

A. Narayanan and V. Shmatikov, "Robust De-anonymization of Large Sparse Datasets," *2008 IEEE Symposium on Security and Privacy*

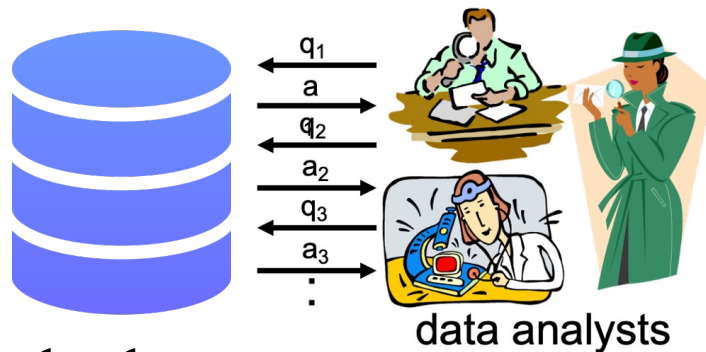
Approach 3: Mediate Access

Name	Sex	Blood		HIV?
Chen	F	B		Y
Jones	M	A		N
Smith	M	O		N
Ross	M	O		Y
Lu	F	A		N
Shah	M	B		Y



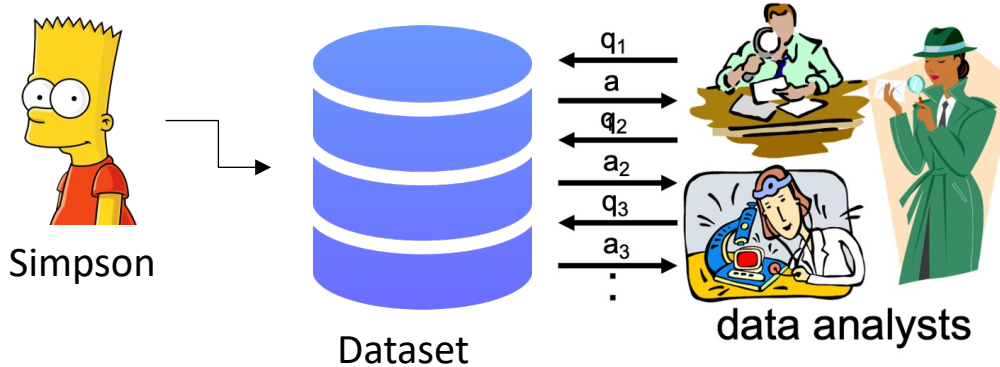
Problems?

Privacy-preserving Data Analysis



- Is the privacy only about
 - Just “De-identification”? No!
 - Just “Statistics”?
 - Such as some quantity calculation
 - Known fact : Overly accurate estimates of too many statistics is blatantly non-private.

How the privacy can be defined?



- What kind of privacy guarantee we want to make to Simpson?

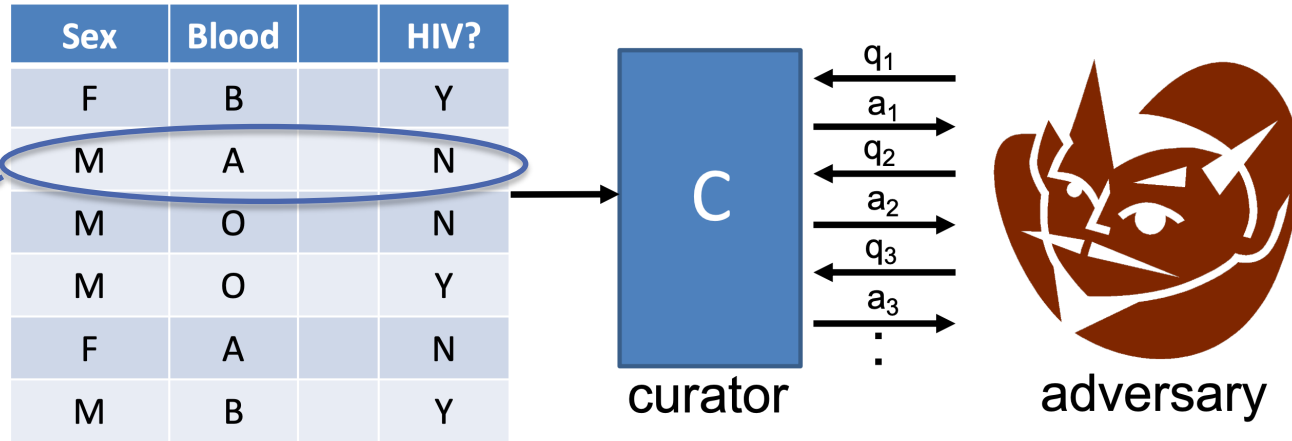
- Can't learn anything new about Simpson?

- ✓ What's is the point?

- Ideally: Learn same things which is independent with any specific individuals. E.g., learn the same thing if Simpson is replaced by Bob (any random people of the population)



Differential Privacy



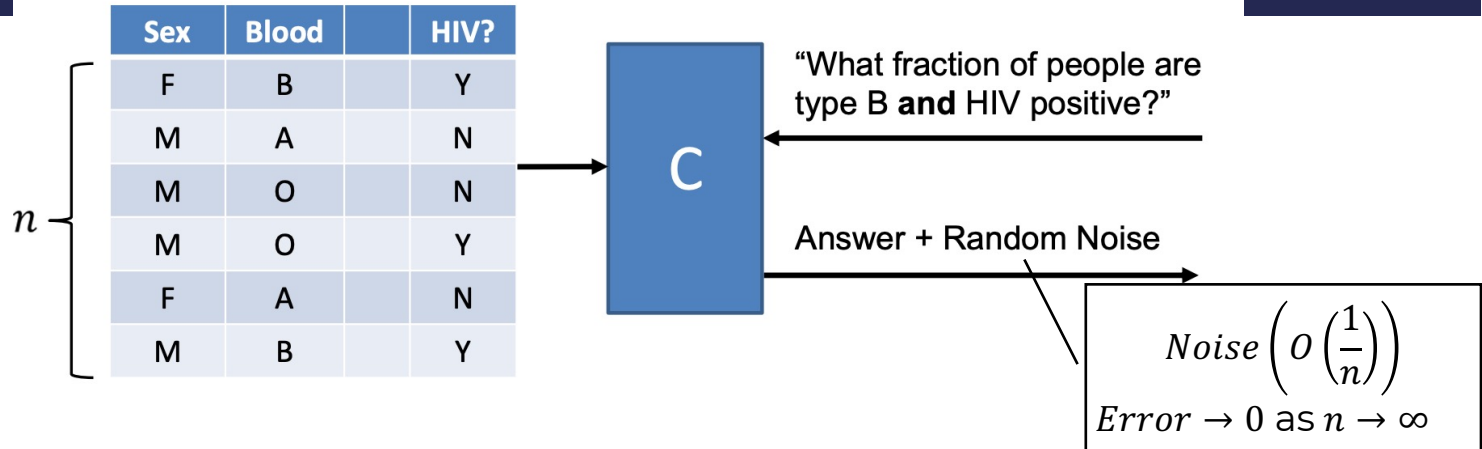
Requirement: effect of each individual should be hidden!

- An adversary shouldn't be able to tell if any one person's data were changed arbitrarily.

Why this definition is reasonable?

- Dropping a user's data will not change the output
- From the output, we cannot tell if a user is in a dataset or not
- In case that you cannot tell whether a user is in or not, you cannot know their data
 - Can protect many attacks to the privacy of users, like re-identification and much more.

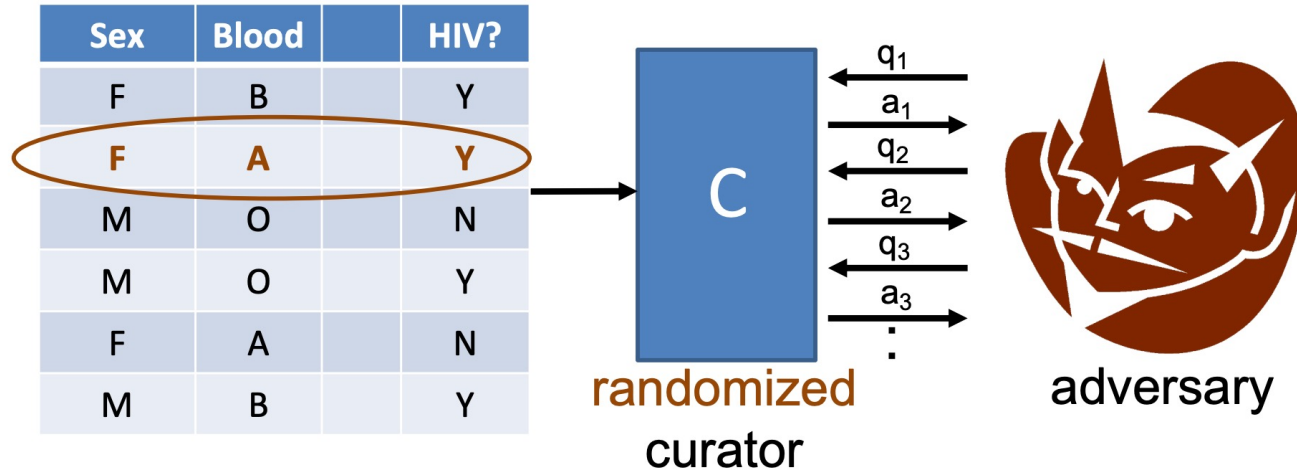
Simple approach: random noise



- Very little noise is needed to hide each person as n is big enough.
- This is just for one query.

Dwork C, McSherry F, Nissim K, et al. Calibrating noise to sensitivity in private data analysis. TCC 2006.

Differential Privacy



Requirement: for all D, D' differing on one row, and all $q_1, \dots, q_t$

Distribution of $C(D, q_1, \dots, q_t) \approx_{\epsilon}$ Distribution of $C(D', q_1, \dots, q_t)$

Formal Definition

- Setup: given n datapoints $X = X_1, X_2, \dots, X_n$ to a trusted curator
- The curator runs an algorithm $M: \mathcal{X}^n \rightarrow \mathcal{Y}$, outputs $M(X)$
- Definition:

An algorithm M is ε -differentially private if for all dataset X and X' which differ in one-entry, and for all $S \subseteq \mathcal{Y}$,

$$\Pr[M(X) \in S] \leq e^\varepsilon \Pr[M(X') \in S]$$

Differential Privacy: Pros and Cons

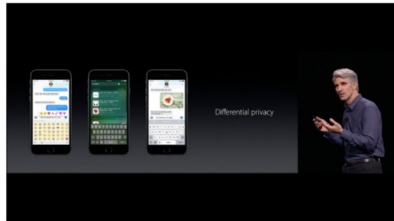
- Pros:

- Mechanism that cannot leak “individual-specific” information
- Composes: k repetitions is $k\varepsilon$ differentially private

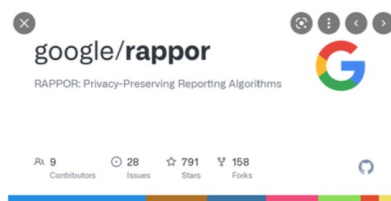
- Cons:

- Cannot protect for information that is not localized to a few rows
- No guarantee that subjects won't be “harmed” by results of analysis.
- Cannot prevent inferences.

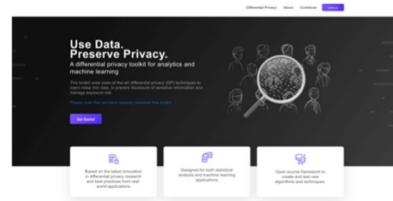
Differential Privacy Deployment



Apple



Google



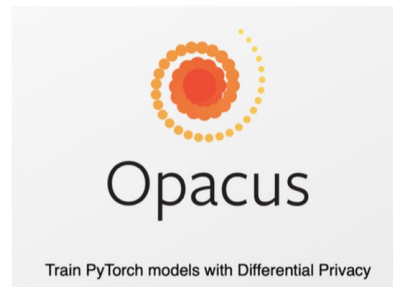
Microsoft



Census Bureau



Uber



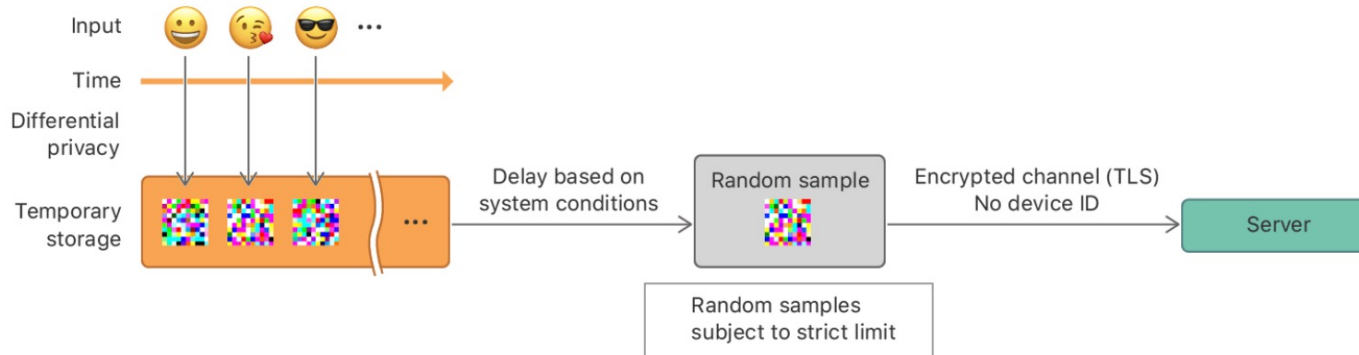
Meta



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Apple: Local Differential Privacy

- Gaining insight into what many Apple users are doing, while helping to preserve the privacy of individual users.



<https://machinelearning.apple.com/research/learning-with-privacy-at-scale>

Differential Privacy: Algorithm Design

- **Answering Numeric Queries:**
 - Global Sensitivity Method: Statistics
- **Answering Non-Numeric Queries:**
 - Exponential Method: optimization

Different Ways of Adding Noise!

- **Problem:**

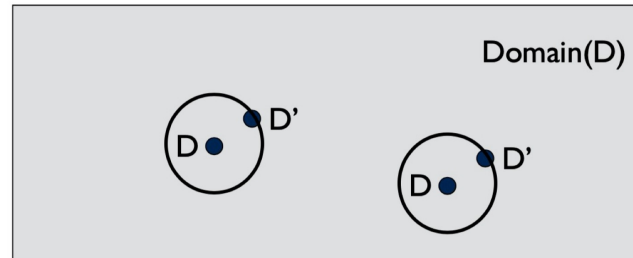
Given function f , sensitive database D ,

To find a differentially private approximation to result $f(D)$

The Global Sensitivity Method

- Assume that dataset D' only differ on 1 item from D , which is obtained by adding or removing one record from D .
- Given a function f and sensitive database D ,
- The global sensitivity of f is defined as

$$S(f) = \max_{\text{dist}(D, D') = 1} |f(D) - f(D')|$$



Intuition: The maximum changes of the function value if you change one item of the function input.

Differential Privacy: The Laplace Mechanism

- **Problem:**

Given function f , sensitive database D ,

To find a differentially private approximation to result $f(D)$

- **Solution** via using global sensitivity of f

➤ Output $f(D) + Z(\text{error})$, where

$$Z \sim \frac{S(f)}{\epsilon} \text{Lap}(0, 1)$$

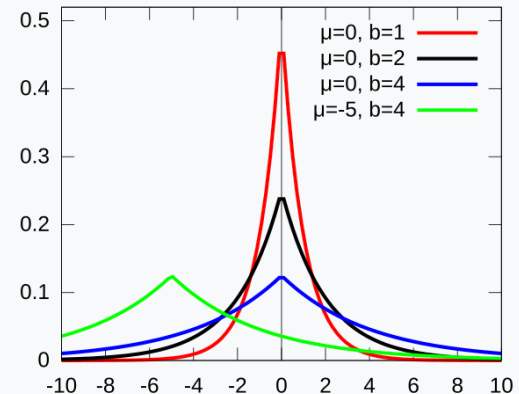
ϵ -differentially
private

Laplace distribution:

$$p(z|\mu, b) = \frac{1}{2b} \exp\left(-\frac{|z - \mu|}{b}\right)$$

Laplace

Probability density function



Example: Calculation Sum on Dataset

- Let $D = \{0,1\}^n$ and $f(D) = \sum_{i=1}^n x_i$
- Then, global sensitivity of f is
$$S(f) = \max |f(D) - f(D')|, \text{ where } D \text{ and } D' \text{ differ on 1 bit.}$$
- Then, $S(f) = 1$
- Output with ε -differentially privacy should be
$$f(D) + Z, \text{ where } Z \sim \frac{1}{\varepsilon} \text{Lap}(0,1)$$

Exponential Mechanism (For Non-numeric Queries)

- Given a “scoring” function $f(w, D)$
- Find w such that the $f(w, D)$ is maximum.

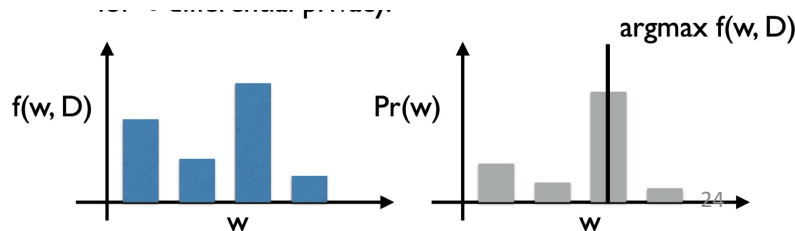
Suppose for any w ,

$$|f(w, D) - f(w, D')| \leq S$$

when D and D' differ in 1 record. Sample w from:

$$p(w) \propto e^{\epsilon f(w, D) / 2S}$$

for ϵ -differential privacy.



Summary

- First presented in 2006, differential privacy is the subject of ongoing research to develop privacy technologies that provide robust protection against a wide range of potential attacks
- It works by adding a small amount of random noise during analytical computations.

Challenges for DP

- Accuracy for “small data” (the size of the number set is small)
- Modeling and managing privacy loss over time

Algorithm Implementation & Resource

- OpenDP Library

<https://docs.opendp.org>

- Materials from Harvard class CS208 by Applied Privacy for Data Science by James Honaker and Salil Vadhan

<https://people.seas.harvard.edu/~salil/cs208/>